Enrolment Number	2	4	1	3	0	0	7	0	3	3

Total No. of printed pages = 03

Winter 2025

B. Tech 2nd Semester Examination

Chemistry – Theory

Course Code: BCH23111T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION 1

1. Answer all the questions **Marks: 10 x 1 =10**

(i) The lowest temperature at which an oil lubricant gives of vapours that ignites for a moment is called

(a) fire point
(b) flash point
(c) refractory point
(d) None of the above

(ii) The entropy change for the fusion of 1 mole of a solid which melts at 300K is (if latent heat of fusion is 2.51 kJ/mol).

(a) 4.37 $kJK^{-1}mol^{-1}$
(b) 83.7 $JK^{-1}mol^{-1}$
(c) 8.37 $JK^{-1}mol^{-1}$
(d) 4.37 $JK^{-1}mol^{-1}$

(iii) Which of the following is **not** a green solvent?

(a) benzene
(b) water
(c) supercritical CO_2
(d) ionic liquid

(iv) Which of the following transition is not possible in UV Visible spectroscopy

(a) $\sigma \rightarrow \sigma^*$
(b)$n \rightarrow \sigma^*$
(c)$\Pi \rightarrow \Pi^*$
(d) $\gamma \rightarrow \Pi$

(v) Proximate analysis of coal involves determination of

(a) Moisture
(b)Volatile matter
(c) Nitrogen
(d) Both (a) and (b)

(vi) An example of primary fuel is

(a)Wood

(b)Coke

(c)Kerosene

(d)Water gas

(vii) Octane number of isooctane is

(a) 0

(b) 100

(c) 10

(d) 40

(viii) During electrochemical corrosion the more active metal acts as

(a) Anode

(b) Cathode

(c) Anode as well as cathode

(d) None of these

(ix) Which of the following is paramagnetic?

(a) N_2

(b) O_2

(c) O_2^{2-}

(d) H_2

(x) Corrosion can be prevented by

(a) alloying

(b) tinning

(c) galvanizing

(d) all of these

SECTION 2

2. Answer one (01) question from this section (Maximum 150 words for each) **Marks: 5**

(a) What is Beer's law? Write any two applications of IR Spectroscopy. **(5)**

(b) Define Gross calorific value and net calorific value of a fuel. What do you mean by octane number and cetane number. **(5)**

SECTION 3

3. Answer any one (01) question from this section (Maximum 300 words for each) **Marks: 10**

(a). Interpret Helmholtz Free energy and Gibbs free energy in thermodynamics mathematically and physically with their physical significances. **(10)**

(b) Explain the twelve principles of green chemistry with suitable examples of each. **(10)**

Tot

SECTION 4

4. Answer any two (02) questions from this section (Maximum 150 words for each) **Marks: 5 x 2 =10**

(a) Compare the criteria of a good lubricant and a good refractory material. **(5)**

(b) 30 moles of an ideal gas at the initial pressure of 1 atmosphere at 0 °C were expanded reversibly under isothermal conditions to a final pressure of 0.1 atmosphere. Calculate the work done by the gas, the change in internal energy and heat absorbed by the system ($R=8.314\ JK^{-1}mol^{-1}$). **(5)**

(c) An electron in a one-dimensional box of length 1 Å undergoes a transition from the ground state (n=1) to the first excited state (n=2). Calculate the transition energy. **(5)**
(Mass of electron = 9.1×10^{-31} kg, h = 6.626×10^{-34} Js)

(d) **(i)** Identify the possible electronic transitions that can happen in the following molecules when they absorb UV-Visible radiation (1) CH_4 (2) CH_3COCH_3 **(2)**

(ii) Identify the no. of signals in the 1H NMR spectrum of the following molecules. **(3)**
(1) CH_3COCH_3 (2) CH_3CH_2OH

SECTION 5

5. Answer any one (01) question from this section (Maximum 150 words for each) **Marks: 5**

(a) Analyse any three characteristics of a good fuel.

(b) Analyse the molecular orbital energy level diagram of N_2 molecule and find out its bond order and magnetic behaviour

SECTION 6

6. Answer any two (02) questions from this section (Maximum 150 words for each) **Marks: 5 x 2 =10**

(a) Determine the gross and net calorific value of a sample of coal having the following compositions:
C=80%, H=7%, 0=3%, S=3.5%, N=2.1 % and ash=4.4% . Latent heat of steam=587cal/g. **(5)**

(b) Discuss the Schrodinger wave equation by elaborating each term involved. **(5)**

(c) Elaborate the different steps of setting and hardening of cement. **(5)**

(d) Discuss waterline and pitting corrosion. **(5)**